BTS –VI(SS)–01.17–0079 Reg. No. | | | | | | | | |

A

B.Tech. Degree VI Semester Special Supplementary Examination January 2017

ME 1604 HEAT AND MASS TRANSFER
(*2012 Scheme*)

Time : 3 Hours Maximum Marks : 100

(Use of approved data book is permitted)

PART A
(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) Explain Fourier's law of conduction.
(b) Explain the lumped heat capacity system.
(c) Differentiate film wise and drop wise condensation.
(d) State Ficks' law of diffusion and give its expression.
(e) What is radiation shape factor? Give any two shape factor algebra.
(f) State Wien's displacement law and Kirchoffs law.
(g) Sketch the temperature variations in parallel flow and counter flow heat exchangers.
(h) What is fouling? Give the fouling factors taken into account in the design of heat exchangers

PART B

(4 × 15 = 60)

II. A pipe was insulated by asbestos and exposed to fluid at 40°C with h = 4.5 W/m^2K. Calculate the critical radius of insulation. Calculate the heat loss from 400°C, 0.07 m diameter pipe covered with insulator with critical radius of insulation and without insulation. Length of the pipe is 2 m.

OR

III. A concrete wall, 20 cm thick, at an initial temperature of 20°C is suddenly exposed on both sides to pure steam at atmospheric pressure. If the thermal resistance of the condensate flowing down the wall is negligible, estimate the rate of steam condensation on 160 m^2 area after (i) 10 s (ii) 3 hours.

IV. Air at 25°C flows past a flat plate at 2.5 m/s The plate measures 600 mm x 300 mm and is maintained at an uniform temperature of 95°C. Calculate the heat loss from the plate, if the air flows parallel to the 600 mm side.

OR

V. A vertical pipe 80 mm diameter and 2 m height is maintained at a constant temperature of 120° C. The pipe is surrounded by still atmospheric air at 30°C. Find heat loss by natural convection.

VI. The sun emits maximum radiation at $\lambda = 0.52\,\mu$. Assuming the sun to be a black body, calculate the surface temperature of the sun. Also calculate the monochromatic emissive power of the sun's surface, total emissive power and maximum emissive power.

OR

VII. A 70 mm thick metal plate with a circular hole of 35 mm diameter along the thickness is maintained at an uniform temperature of 250°C. Find the loss of energy to the surroundings at 27°C, assuming the two ends of the hole to be as parallel discs and the metallic surfaces and surroundings have black body characteristics.

VIII. Saturated steam at 120°C is condensing on the outer tube surface of a single pass heat exchanger. The heat transfer coefficient is $U_o = 1800\ w/m^2K$. Determine the surface area of a heat exchanger capable of heating 1000 kg/hr of water from 20°C to 90°C. Also compute the rate of condensation of steam $h_{fg} = 2200 kJ/kg$.

OR

IX. A counter flow, concentric tube heat exchanger is used to cool the lubricating oil $(C_p = 2.45\,kJ/kg^oC)$ for a large industrial gas turbine engine. The flow rate of cooling water $(C_p = 4.2\ kJ/kg^oC)$ through the inner tube $(D_i = 25\ mm)$ is 0.2 kg/s, while the flow rate of oil through the outer annulus (D_o = 45 mm) is 0.1 kg/s. The oil and water enter at temperatures of 100°C and 30°C, respectively. Determine the length of the tube if the outlet temperature of the oil is to be 60°C. What would be the length, if the fluid flow were parallel?

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